

Teaching Exam

**Subject –
Computer
Science**

**COMPUTATIONAL THINKING
AND PROGRAMMING**

Lecture No –5



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Topics to be Covered



Topic

list
Tuples
Strings

Dictionary → conditional
Statements
Q
Loops

list.append. (✓)

list.extend.

list.sort.()

sorted ()

list.reverse()

a = ['list', 'list 1', 'list 11']

list.insert(—, —)

list.remove(^{element} —) ⇒ # remove element of present

list.pop(idx) # remove element of particular idx

list

Tuple

[0 1 2 3 4 5]
 ↓ ↓ ↓
 a b

Dictionary [key = value.]

dict { 'a' = 'value 1',
 'b' = 2 + 5j,
 'c' = 18,
 'd' = 'value 2' }

string
 list
dict
tuple

← ['d'] = 'value 2' }
 key value

int, string, tuple



dict.get("key") # ~~it~~ returns corresponding element of key

dict['a'] $\Rightarrow$ 'c' :

dict.items() # returns all keys and values.
in tuple

flow of statements

conditional
Statements

if, ^{else}, else, else.

Loop

↳ while (-)

↳ for (sequential)

if (true)

→ if (true)

if (true)
 ~~else~~
 else -
 else

[~~~~~]

if :

if (1 > 0)

=====

input $\rightarrow$ age.

age > 18

age ^{driving} $< \underline{80}$ (driving licence can be owned.)

can drive.

cannot drive.

~~int~~ age = ^{int} (input (age))

if (age > 18 and age < 80):

--- print ("can drive".)

else:

print ("cannot drive")

input 3 no. - greater of all 3 no.

a, b, c.



if ($a > b$ and $a > c$)

print (a is largest.)

elif ($b > c$)

print (b is second largest.)

else

print (c is third largest.)

if ($b > c$)

print (b is greatest.)

if ($a > c$)

else

print (c is greatest.)

if ($b > a$)

a, b, c

after sum (ascending order.)

lowest ————— greatest.

if (a > b) $\Rightarrow$ X

a, b = b, a.

if (a > c)

a, c = c, a.

if (b > c.)

b, c = c, b

print [a, b, c]

5, 12, 7.

5, 7, 12.

15, 18, 7.

15, 7, 18

7 18 15

7 15 18



2 mins Summary



TEACHING
WALLAH

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THANK YOU